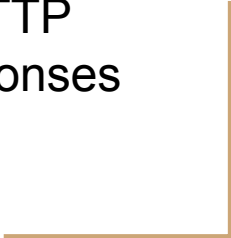




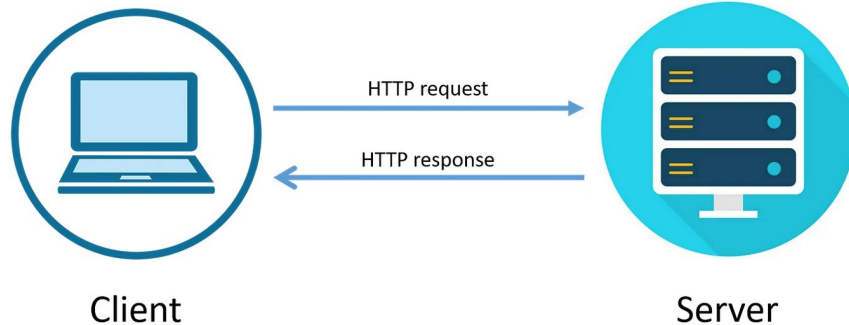
HTTP Request Protocol

Understanding HTTP
Requests and Responses



Introduction to HTTP

- HTTP (Hypertext Transfer Protocol) is the foundation of data communication on the web.
- It is a request-response protocol between a client (browser) and a server.
- HTTP is stateless, meaning each request is independent of others.



HTTP Request Methods

- **GET:** Retrieve data from the server.
- **POST:** Send data to the server to create a resource.
- **DELETE:** Remove a resource from the server.
- **PATCH:** Partially update a resource on the server.
- **PUT:** Replace a resource on the server.
- **HEAD:** Retrieve headers without the body.
- **OPTIONS:** Describe communication options for the target resource.
- **CONNECT:** Establish a tunnel to the server (used for HTTPS).
- **TRACE:** Echoes the received request for debugging purposes.

Anatomy of an HTTP Request

An HTTP Request consists of:

1. **Request Line:**

- Method (e.g., GET, POST)
- URL (e.g., [/index.html](#))
- HTTP Version (e.g., [HTTP/1.1](#))

2. **Headers:**

- Key-value pairs providing metadata (e.g., [Content-Type](#), [Authorization](#)).

3. **Body (optional):**

- Data sent to the server (e.g., JSON, form data)

Example of an HTTP Request

```
GET https://jsonplaceholder.typicode.com/albums/1/photos HTTP/2
```

```
Host: jsonplaceholder.typicode.com
```

```
Accept-Language: en-US
```

```
Cache-Control: no-cache
```

- **Request Line:** GET jsonplaceholder.typicode.com/albums/1/photos HTTP/2
- **Headers:** Host, Accept-Language, Cache-Control
- **Body:** No body sent in this example

Anatomy of an HTTP Response

An HTTP Response consists of:

1. **Status Line:**

- HTTP Version (e.g., HTTP/2)
- Status Code (e.g., 200)
- Status Message (e.g., OK)

2. **Headers:**

- Key-value pairs providing metadata (e.g., Content-Type, Server).

3. **Body (optional):**

- Data returned by the server (e.g., HTML, JSON).

Example of an HTTP Response

HTTP/2 200 OK

Content-Type: application/json

Server: cloudflare

Content-Length: 1820

```
[  
  {"albumId":1,"id":1,"title":"...","url":"...","thumbnailUrl": "..."},  
  ...  
]
```

- **Status Line:** HTTP/2 200 OK
- **Headers:** Content-Type, Server, Content-Length
- **Body:** JSON data on all albums with photos

HTTP Status Codes

- **1xx (Informational):** Request received, continuing process.
- **2xx (Success):** Request successfully received, understood, and accepted.
 - 200 OK, 201 Created, 204 No Content
- **3xx (Redirection):** Further action needed to complete the request.
 - 301 Moved Permanently, 304 Not Modified
- **4xx (Client Error):** Request contains bad syntax or cannot be fulfilled.
 - 400 Bad Request, 404 Not Found, 403 Forbidden
- **5xx (Server Error):** Server failed to fulfill a valid request.
 - 500 Internal Server Error, 503 Service Unavailable

Common HTTP Headers

- **Request Headers:**

- **Host**: Domain name of the server.
- **User-Agent**: Client software details.
- **Authorization**: Credentials for authentication.
- **Content-Type**: Type of data in the body (e.g., `application/json`).

- **Response Headers:**

- **Content-Type**: Type of data in the body.
- **Server**: Software used by the server.
- **Set-Cookie**: Sets a cookie in the client.
- **Cache-Control**: Cache the data for optimization

Useful links

- [HTTP Request Methods](#) (MDN Docs)
- [HTTP Response Status Codes](#) (MDN Docs)